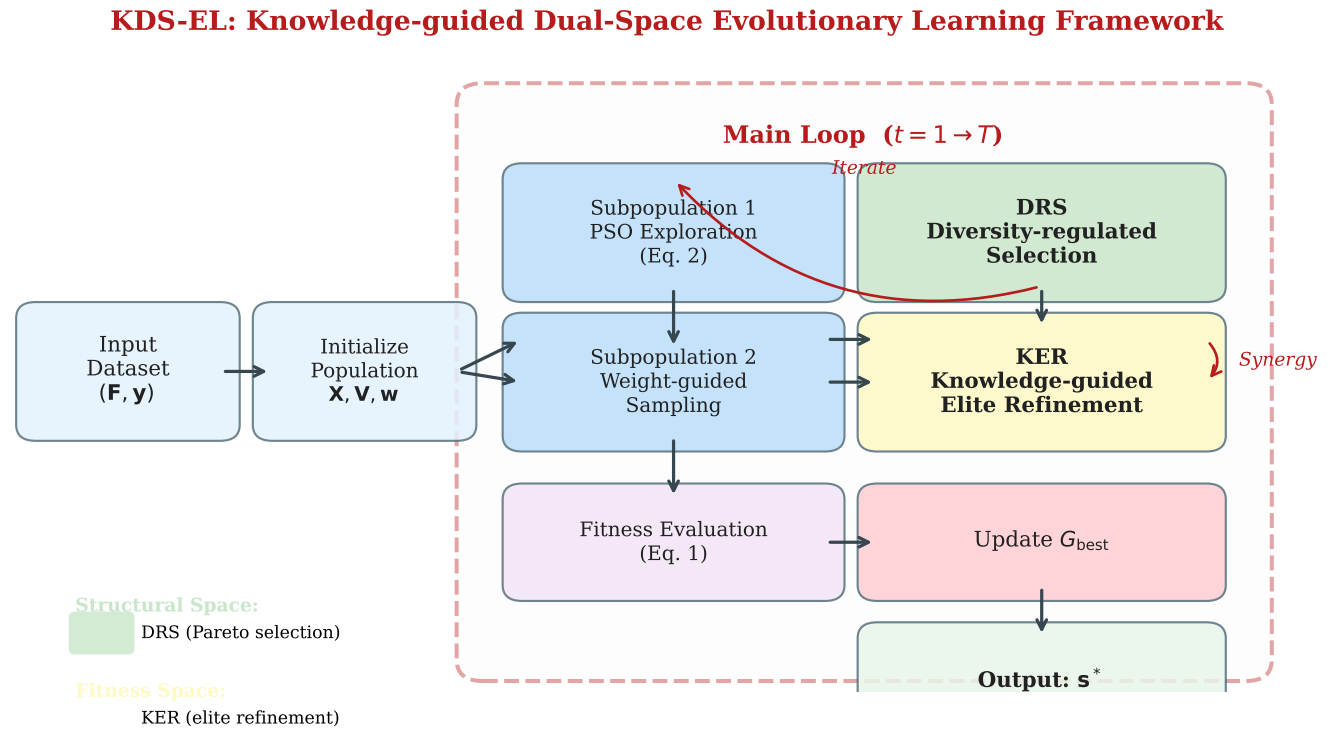
KDS-EL: Knowledge-guided Dual-Space Evolutionary Learning Framework
Main Loop ($t = 1 \to T$)
Iterate
Input Dataset ($\mathbf{F}, \mathbf{y}$)
Initialize Population $\mathbf{X}, \mathbf{V}, \mathbf{w}$
Subpopulation 1 PSO Exploration (Eq. 2)
Subpopulation 2 Weight-guided Sampling
DRS Diversity-regulated Selection
KER Knowledge-guided Elite Refinement
Synergy
Fitness Evaluation (Eq. 1)
Update G_{best}
Output: $\mathbf{s}^*$
Structural Space:
DRS (Pareto selection)
Fitness Space:
KER (elite refinement)